

Not every stable cluster is homeostatic: Stability, network order, and control in projectible kinds

Brett Reynolds *

Humber Polytechnic & University of Toronto

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Abstract

Homeostatic property cluster theory gave natural-kinds realism a powerful anti-essentialist form, on which a kind can be real because its properties cluster through mechanisms rather than because it has an essence. The word HOMEOSTATIC has tended to carry more weight than the theory has earned, because it is used for several achievements that are not the same: reliable co-occurrence, network order, maintenance, and feedback-sensitive control. This paper treats projectibility as the target of kind claims and offers a diagnostic ladder: identify the projection target; ask whether a stable profile supports it; ask whether network order makes the profile non-accidental; ask what, if anything, maintains the profile; and ask whether that maintenance is corrective control. Homeostasis belongs only at the last step, where perturbation-sensitive organization preserves a higher-scale relation through lower-scale variation. The Watt governor calibrates control, glycemic regulation tests strict biological control, interjections supply a negative linguistic case, and agreement monitoring supplies a candidate positive one. The result is a disciplined way of using the framework's insight without turning HOMEOSTATIC into a loose synonym for stable or projectible.

Keywords: natural kinds; homeostatic property clusters; projectibility; homeostasis; causal control; linguistic categories

I INTRODUCTION

The target of this paper is a quiet ambiguity in recent anti-essentialist kind theory. Boyd's homeostatic property cluster framework, Slater's stable property cluster proposal, and Khalidi's causal-network account all reject essences, and they all aim to explain why some categories support projection while others do not. Each, though, locates the basis for projection somewhere slightly different.

Boyd ties it to homeostatic mechanisms that maintain the cluster (Boyd, 1991); Slater deliberately sets the stability of a cluster apart from the means by which it is maintained (Slater, 2015); and Khalidi locates it in the causal-network order among a kind's properties (Khalidi, 2013, 2018). The ambiguity

*Contact: brett.reynolds@humber.ca

lies less in these accounts than in the shared vocabulary of “homeostatic property clusters”, which covers all of these options at once and so leaves underdetermined what secures projection in any given case.

The looseness is clearest in the homeostatic-mechanism component of the framework. Boyd gives that component considerable flexibility, treating the clustering as “metaphorically (sometimes literally)” homeostatic and allowing it to be sustained by mutual property support, underlying mechanisms, or both (Boyd, 1999). Onishi and Serpico are therefore right that the notion of a homeostatic mechanism in HPC theory has never been fully explicated (Onishi & Serpico, 2022).

The claim is that not every stable cluster is homeostatic. The literature need not explicitly identify every stable cluster with a homeostatic one for this to matter. Shared HPC vocabulary still makes it too easy to slide from stable profile to stabilizer, from stabilizer to homeostasis, and from homeostasis to projectible kind, as though these were one achievement.

The order should run the other way. Start from a projection target, ask what empirical profile supports it, ask what network order makes the profile non-accidental, ask what maintains it, and only then ask whether the maintenance is corrective control. This order keeps Boyd’s projectibility frame in place while asking that HOMEOSTASIS be used more literally, in a way that makes a genuine demand. Weinberger’s recent account of homeostasis as a form of causal control is what supplies that demand (Weinberger, 2026).

The contribution is therefore diagnostic rather than revisionary. It leaves HPC theory in place and separates five questions that the shared vocabulary tends to compress. What is the projection target? What stable profile supports it? What network order, if any, explains why the profile is not an accidental list? What maintains that profile across the relevant range of cases? Does the maintaining process correct perturbations and thereby preserve a higher-scale relation? Only this last question gives the word HOMEOSTATIC anything to mean beyond stability.

The paper’s claim is not that HPC theory should become control theory. It is that HPC theorists need a diagnostic for when control is actually present.

The scope is Boyd-style projectible realist kind claims. Many classifications earn their keep by mapping variation, stabilizing measurement, organizing historical comparison, or coordinating inquiry before they support strong projection. Those uses are legitimate, but the diagnostic here concerns categories offered as real kinds because they support prediction, explanation, induction, or controlled intervention.

The cases that follow run across a machine, an organism, and two linguistic categories. They are not offered as a survey of those domains. They are chosen because they stress different rungs of the ladder and test whether a single, domain-neutral diagnostic can keep projection, stable profile, network order, maintenance, and corrective control apart. A distinction in the theory of kinds ought to travel in this way, and if it held only in biology that would be a limitation rather than a sign of rigour. The cases earn their place by what they reveal about the diagnostic, not by representing their fields.

The order of what follows is straightforward. The next five sections develop projectibility and the successive evidential demands, ending with corrective control. A short section then states the diagnostic ladder before the remaining sections work through the cases and ask what the distinction changes for HPC theory more broadly.

2 PROJECTIBILITY FIRST

Boyd's framework works because projectibility does not require necessary and sufficient conditions. A category can support induction when its associated properties are connected non-accidentally, even where no single property is shared by every member. On Boyd's accommodation thesis, successful classification is a matter of fitting our inferential practices to causal structure, so that induction and explanation "always require that we accommodate our categories to the causal structure of the world" (Boyd, 1991, p. 224). We construct the categories, but whether they succeed depends on a structure we do not control, and it is that structure which does the explanatory work.

The problem of PROJECTIBILITY comes from Goodman's new riddle of induction, where a predicate can fit all observed cases and still fail to license projection to unobserved ones (Goodman, 1955/1983). Boyd's contribution was to ground projectibility in causal structure without grounding it in essences, which is also what gave HPC theory its purchase against the worry, which Boyd associated with Hacking, that where a kind's boundaries are fuzzy its membership is fixed by "people in concert" rather than by nature (Boyd, 1991).

This is one reason it helps to begin from the projection target rather than from mechanism. Mechanisms matter to the theory because they explain why projection is reliable, so a category that supports no projection is not redeemed by having a tidy causal story behind it, and a category that does support projection is already doing much of the work a realist about kinds cares about. The task of the rest of the paper is to ask, for categories that are clearly projectible, which further evidential demands they meet.

The risk is that this question can be made too easy. If the only thing asked is whether a cluster recurs, a great many categories will qualify, because recurrence on its own is cheap. Stable co-occurrence may license some inference, but an analysis still has to specify what is being projected from what before the category becomes theoretically interesting, and that happens only when its profile supports prediction, explanation, or controlled intervention.

3 OBSERVED STABILITY AND INVARIANCE

Slater's stable property cluster account presses the useful point that homeostatic mechanisms are not always needed for natural kindness. What matters on this view is the stability of a property cluster, considered apart from the various ways that stability might be maintained, since a cluster can, in Boyd's phrase, hold itself together (Slater, 2015). This yields the first distinction the paper needs, because a stable and projectible cluster may be perfectly real without being homeostatic in any stronger sense.

Once the target is fixed, the next question is whether a stable profile supports it. Stability that is left unexplained is nonetheless too thin to do all the work, and Onishi and Serpico turn this into a dilemma.

If the explanatory role of underlying processes is made optional or postponed, as stable-cluster views invite us to do, the account risks leaving underexplained why some inductions are rational rather than lucky. If causal structure is retained through causal-network order, as Khalidi proposes, the analysis still faces interest-relativity, because which structure counts as the relevant one depends on the explanatory purpose at hand (Onishi & Serpico, 2022). Their broader observation is that the notion of a homeostatic mechanism has never been fully explicated, and that interest-relativity is unlikely to be eliminable.

The diagnostic offered here does not try to remove that interest-relativity so much as to make it explicit. Asking first what a category lets us project, and against which declared expectation, fixes the explanatory target, and the question of securing structure is then asked relative to that target rather than in the abstract. Interest-relativity becomes a parameter the analysis declares rather than a cost it pays without noticing.

Woodward's account of invariance helps locate what mere stability leaves out. The kind of stability that matters for explanation is invariance under intervention, not simply persistence across space and time, so that a generalization earns its explanatory keep when it would continue to hold across a relevant range of changes, including changes that could be imposed (Woodward, 2001). The first diagnostic step after the projection target is correspondingly modest: a stable property cluster should support some projection across a specified range of contexts and interventions, with evidence that goes beyond recurrence and shows that observing part of the profile licenses a useful expectation about another part, or about how the system behaves under a relevant change.

4 NETWORK ORDER WITHOUT CONTROL

Khalidi's network account addresses a different shortcoming. A cluster is more than a list of associated properties, because its properties can be ordered, with some generating, sustaining, constraining, or explaining others. This is an advance over a bare cluster picture, since it explains why some properties are diagnostic and others merely derivative (Khalidi, 2013, 2018). Khalidi himself replaced the notion of a homeostatic mechanism with causal-network order in pursuit of a more general account, so identifying that order as one securing structure among several is continuous with his aim.

That order is the strength of the account. The present point is narrower: ordered dependence, even when objective and explanatory, does not by itself amount to feedback-sensitive correction.

Ordered dependence is still some distance from corrective control. A node in a causal network can support projection because the properties around it stand in systematic causal relations, and that is compatible with the system never tracking a departure, never allowing lower-scale variables to adjust, and never preserving a higher-scale relation through any such adjustment. Network order can therefore explain projectibility without amounting to homeostasis.

The network-order step accordingly asks whether the profile has internal structure, and the relevant evidence consists of asymmetries among the properties: which are upstream, which derivative, which central, and which only peripheral cues. The characteristic failure is the use of causal vocabulary for what is really an unordered list, since redescribing co-occurrence in causal terms does not by itself establish that any ordering obtains.

I use NETWORK ORDER broadly across the diagnostic. In strictly biological or mechanical cases the ordering will usually be causal, whereas in linguistic and social cases the relevant dependencies may instead be functional, conventional, sequential, or evidential. What matters in each case is that the profile has a structure that explains projection, because a merely associated bundle of features is not enough.

The broad use still needs a constraint. A non-causal ordering relation should be asymmetric: one feature conditions, licenses, constrains, or structures another. Evidential usefulness alone is not network order. A cue is evidentially privileged only when it tracks an ordering relation in the profile rather than merely co-occurring with the rest of the bundle.

5 FROM STABILIZERS TO CONTROLLERS

It helps to separate two questions that talk of maintenance tends to run together. Whatever keeps a cluster clustered can be called its **STABILIZER**, in the sense that removing it would let the category fragment, drift, or dissolve. The stabilizer is a functional role, and the causal structure that fills the role is the mechanism. This is the functional-role description of what Boyd called a homeostatic mechanism (Boyd, 1991), and it has become a familiar way of talking about how linguistic and social categories are held together. Maintenance, in this sense, is the umbrella term.

The question the paper presses is which stabilizers are also controllers. A **CONTROLLER** is the narrower case: a stabilizer with feedback-sensitive corrective organization that preserves a higher-scale relation through lower-scale variation. Not every stabilizer meets this condition. Reliable co-occurrence can be maintained by entrenchment, by transmission, or by a frozen accident that nothing actively corrects, and ordered dependence can be maintained by causal wiring that carries no error signal and returns the system to no target.

The two ideas answer different counterfactuals, which is why they come apart. The stabilizer question is about the maintainer, and it asks what would happen to the cluster if the maintaining process were removed. The controller question is about the profile under threat, and it asks whether a perturbation to the profile is sensed and answered in a way that brings the profile back toward a relation worth projecting from.

A cluster can answer the first question and fail the second, since entrenchment and transmission may hold a profile together without any of them detecting or correcting a departure. The space between a profile that is merely held in place and one that is actively corrected is where **HOMEOSTATIC** tends to be overclaimed.

Read this way, maintenance is the fourth question in the ladder, not the first. A stable profile can support projection before its maintainer is known, and a network-ordered profile can explain why projection is reliable without yet showing what keeps the network in place. Once a maintainer is identified, the final question is whether that maintainer is a controller. Only the controller variety earns **HOMEOSTATIC** in anything like a literal sense, and the next section says what control adds, using the account that states the standard most carefully.

6 CORRECTIVE CONTROL

Control-secured kinds are more demanding, and the first thing to get right is the level at which the control is located. Kinds do not themselves sense, compare, or correct anything. A kind earns the label only when its projectible profile is maintained by a system or a practice that has feedback-sensitive corrective organization, so that the bearer of the control is the system rather than the category named.

Weinberger's central lesson is that homeostasis requires more than constancy, because a controlled system can preserve a higher-scale relation precisely by letting its lower-scale variables move. His dynamic causal models show that feedback systems, and the Watt governor in particular, produce control relations that static causal models and a simple vocabulary of intervention tend to obscure (Weinberger, 2026).

Two consequences are worth drawing out. The first is that holding a variable rigidly fixed can be a sign of less control rather than more, since appropriate variation in the lower-scale variables is often the means by which the higher-scale relation is kept in place. The second is that not every intervention is equally probative. Perturbations to workload or demand can reveal the controller's response, while

clamping the regulated variable itself can sever the very feedback that constitutes the control. Stillness under a clamp is therefore not evidence that nothing was being controlled.

For the purposes of this paper, corrective control has four evidential marks. There must be a perturbation that threatens the projectible profile; a variable or relation that the system tracks and that matters to the profile; a response pathway that changes lower-scale conditions; and a result in which that response preserves a higher-scale relation supporting projection. Feedback on its own does not satisfy these conditions, because what has to be shown is that the feedback explains why the profile remains projectible when it is perturbed.

The biological control literature makes the same warning concrete. Control mechanisms sense conditions, transmit signals, and alter constraints within the mechanisms they control (Bich & Bechtel, 2022a), and in organisms these controls are integrated across the system rather than issued from a single command point (Bich & Bechtel, 2022b; Bich et al., 2016). This is a good deal more specific than the claim that a property cluster is stable, or that its properties stand in some causal order, and it is the final and most demanding step in the diagnostic ladder.

7 THE DIAGNOSTIC LADDER

The proposal is best read as a diagnostic ladder rather than as an exhaustive taxonomy. Control-secured cases will usually display stability, network order, and maintenance as well, but the inference does not run in reverse. A stable cluster need not have an internally ordered profile, a network-ordered kind need not have a specified maintainer, and a maintained kind need not be maintained by corrective control.

The ladder isolates five questions that are recurrently compressed in HPC discussions: what is projected, what profile supports the projection, what order makes the profile non-accidental, what maintains the profile, and whether that maintenance is corrective control. Selection, drift, institutional scaffolding, exemplar effects, and path dependence may each stabilize projectible profiles in ways that deserve separate treatment. These cases matter because they may secure projectibility without fitting neatly into the strict control role; the ladder is meant to prevent conflation, not to classify every stabilizing process.

The ladder is meant to turn the question around. For any proposed HPC kind, it asks five things before the word *HOMEOSTATIC* is used. The first is what the category lets us project. The second is what stable profile supports that projection. The third is whether the profile has ordering relations that explain why the projection is not accidental. The fourth is what maintains the profile. The fifth is whether the maintainer corrects perturbations. The aim is not to police a word but to make the evidential demand behind it explicit.

Nothing in the argument requires every historical use of *HOMEOSTATIC* in HPC theory to have meant feedback control. The claim is normative and diagnostic: corrective control is the structure that gives the term distinctive content once stability, maintenance, and order have been separated.

8 HOW TO READ THE CASES

The cases that follow do not all have the same standing, and the differences are part of the point. The Watt governor is not put forward as a kind at all; it serves to calibrate the control structure that a homeostatic-control-secured kind would have to involve. Glycemic regulation is the strict biological control case, though one in which the familiar set-point picture is itself under pressure. Interjections

Diagnostic step	Evidence needed	Failure mode
Projection target	A specified inference, prediction, explanation, or intervention the category is supposed to support	Kind talk without saying what the category lets us project
Stable profile	Reliable co-occurrence across relevant contexts, with a projection from observed to unobserved features	Mere recurrence without projectible payoff
Network order	Asymmetries among properties, with some upstream, central, derivative, or evidentially privileged	Causal vocabulary used for an unordered list
Maintenance	A stabilizer whose removal would make the profile fragment, drift, or dissolve	Treating persistence as if a maintainer had been specified
Corrective control	Perturbation-sensitive correction that preserves a higher-scale relation through lower-scale variation	Treating stability, maintenance, or feedback as control

Table 1: A diagnostic ladder for projectible profiles and homeostatic control.

are a linguistic category whose stable projectibility and partial network order are clear enough, and they test why those rungs do not yet earn the stronger label. Agreement monitoring is a candidate positive linguistic case, where the stronger label comes closest to being earned once the relevant repair loop is specified.

They are not all kinds in the same sense. They are arranged around the evidential structure required for kind claims: whether a profile supports projection, what order explains that projection, what maintains the profile, and whether the maintainer is corrective control. The governor calibrates the final question; the other cases test how far biological and linguistic profiles move up the ladder.

Running these cases across a machine, an organism, and two linguistic categories is deliberate. The diagnostic concerns how projectibility is secured, and that is not a question internal to any one science. If the same ladder discriminates among a governor, a regulatory profile, and two kinds of linguistic patterning, then the distinction is doing work in the theory of kinds rather than reporting the folklore of a particular field.

9 CALIBRATION CASE: WATT GOVERNOR

The Watt governor is a useful calibration case because its control structure is explicit. A change in engine speed moves the governor's flyballs; their position changes the valve aperture; the aperture changes the flow of steam; and the steam flow changes the speed of the shaft. The controlled target is shaft speed, mediated by the admission of steam, and a clamp placed directly on shaft speed would sever the incoming causal relations and destroy the control relation altogether (Weinberger, 2026).

That point does not make the governor immune to intervention. A change in workload or demand is the right kind of perturbation, because it lets the feedback relation operate and reveals whether lower-scale adjustment preserves the higher-scale coupling.

The governor shows clearly why HOMEOSTATIC ought not to mean constant. Its lower-scale variables are allowed to vary precisely so that workload and steam flow remain coupled in a stable

higher-scale relation. This is the pattern the diagnostic uses to calibrate its final rung, in which a perturbation is met by responsive adjustment in the lower-scale variables, and a relation that supports projection is preserved through that adjustment.

Because the governor is an artifact with a target fixed by its designer, it shows the control structure in an especially clean form, and that is what makes it a reference point rather than a kind. A candidate kind can then be tested against it, and the question for any proposed homeostatic kind becomes whether it exhibits this structure, rather than whether its properties happen to stay put.

10 BIOLOGICAL EXTENSION: GLYCEMIC REGULATION

Glycemic regulation is included not to show that biological homeostatic cases are easy, but to show why the control diagnostic should be relation-preserving rather than set-point-preserving. The relevant candidate is not glucose as a substance, nor a single blood value, but the normoglycemic regulatory profile of an organism. The standard endocrine picture already supplies a control architecture, with a device that sets a normal range, a sensor that measures the regulated variable, a detector that compares the measured value against the range and emits an error signal, a controller, and an effector that moves the variable back toward the range (Bich et al., 2020). Taken at face value, this looks control-secured.

The projectible payoff is the organism's metabolic response across changing conditions. Given the normoglycemic regulatory profile, one expects coordinated changes in insulin, glucagon, hepatic output, and glucose uptake across feeding, fasting, and activity, rather than an isolated reading that happens to fall within a range.

The standard model already reserves the word **CONTROLLER** for the component that reads the error signal and drives the effector. The vocabulary of the calibration case and of the biological case coincides, and not by accident, since both are feedback architectures in which a sensed departure drives a correcting response. The question is never whether the word fits the diagram, but whether the organism realizes the architecture that the diagram describes.

Two complications make that question sharper. The first concerns the set point, since population correlations for glucose and calcium may fit an equilibrium model better than they fit a defended set point (Fitzgerald & Bean, 2018). The second is that Bich et al. (2020) argue the feedback-loop diagram flattens a hierarchical organism into a chain of components, encourages a search for neatly localized parts, and assumes a normal range without explaining what establishes it. They move instead to organizational closure, on which glucose regulation is part of the organism's self-maintenance rather than the tracking of an externally given number. Read this way, the corrective structure is real, but the set point is the element the cybernetic analogy gets wrong.

That move marks the real contrast with the governor. The governor's target is set from outside by a designer, whereas the organism's range is constituted by what keeps the organism alive. The higher-scale relation that glycemic control preserves is therefore not a number but a relation between glucose availability and the organism's continued functioning, held in place through wide variation in the lower-scale variables across feeding, fasting, and activity. A biological example counts as homeostatic-control-secured only when the evidence displays this organized correction, and not when the variable merely stays within range.

Glycemic regulation is therefore the strict biological case in the paper: the stable profile is maintained by organized responses to perturbation, and what the system returns to is best understood as the relation rather than the reading.

11 CONTRAST CASE: INTERJECTIONS

English interjections are the negative linguistic case. The projection target is pragmatic and interactional: identifying a form as an interjection licenses expectations about how it will be packaged, placed, and taken up in use. The stable profile is familiar, since prosodic isolation, low syntactic integration, non-inflection, non-referentiality, and interactional force tend to travel together (Ameka, 1992; Norrick, 2009). Observing some of these properties licenses expectations about the others, which makes the profile projectible rather than a mere list.

The payoff is pragmatic. Forms such as *huh?*, *mmhm*, and *oh* support expectations about sequential position and interactional uptake, including repair initiation, continuation, and a change in epistemic state (Dingemanse, 2024), while forms such as *wow*, *oops*, *hey*, *well*, and *uh* invite expectations about stance, event management, attention, transition, or delay. Norrick (2009) is helpful here because he treats interjections as an open-ended class with recurrent pragmatic-marker functions.

The contrast between *oh* and *uh* shows the kind of projection at issue. In an appropriate sequential position, *oh* leads us to expect a change-of-state uptake: the speaker has just registered, noticed, or reclassified something. *uh*, by contrast, leads us to expect delay or floor-holding rather than a completed uptake. These are genuine projections from form and placement to interactional role, but they do not yet show a controller preserving the category profile.

The list is deliberately heterogeneous, since several of these forms are analysed as discourse markers, fillers, summonses, or response tokens rather than as central interjections. This heterogeneity is part of the test, because the contrast case is not a classical kind with settled boundaries but a projectible interactional profile whose kindhood is itself partly at issue.

The likely stabilizers include conventionalization, turn-taking routines, prosodic recognizability, acquisition, register, uptake, and metapragmatic learning, and together these make the category stable and useful. They also give parts of the profile some network order, since sequential position, prosodic packaging, and syntactic non-integration are not wholly independent of one another. That gets interjections several rungs up the ladder. It does not yet put them at the top.

The homeostasis question is more demanding. For interjections, the higher-scale relation would be an interactional profile linking turn position, prosodic packaging, low integration, and uptake. The lower-scale variables would include token form, intonation, sequential placement, phonological reduction, and local conventional meaning, and the relevant perturbations would include syntactic integration, inflection, shifted prosody, misplaced sequential use, or failure of uptake. A homeostatic-control claim would require evidence that perturbations of this sort reliably trigger coordinated correction that preserves the higher-scale interactional relation. Repair, failures of uptake, teaching, and metalinguistic policing may sometimes furnish control-like evidence, but the evidence has to be produced, and until it is, interjections belong under the stable-profile, network-order, or maintenance rungs.

The case is useful because it shows how far stable pragmatic projectibility can go before a control claim has been earned.

12 CANDIDATE POSITIVE CASE: AGREEMENT MONITORING

Agreement monitoring is the linguistic case where the stronger label comes closest to being earned. In agreement systems, one expression fixes features that other expressions must express. In Corbett's terms, the source is the controller, the agreeing expressions are targets, and canonical agreement is

the case in which the features take matching rather than mismatching values (Corbett, 2006). The features may be number, person, gender, noun class, or case. Gender is therefore one possible feature in the system, not the object of the case.

That terminology is useful but not decisive. Corbett's controller-target relation identifies a dependency structure in agreement. By itself, it is not a feedback-control architecture. The controller does not sense a mismatch, compare target forms with a standard, or issue a corrective signal. Agreement morphology establishes the relation whose preservation matters; production monitoring and self-repair are where a control-theoretic claim has to look for feedback.

The projectible payoff is clear. Once the controller's features and the agreement domain are known, target forms can be projected across the domain. Number agreement lets us project verb form from subject number. Noun-class agreement lets us project class-marking across verbs, adjectives, and demonstratives (Ashton, 1947; Corbett, 2006). The dependency is also ordered rather than merely stable, because the controller-target relation is asymmetric: the controller fixes the relevant feature values, and targets express them.

The first three rungs are therefore straightforward: agreement has a projection target, a stable profile, and an ordered dependency relation. The harder question is whether any maintenance process for that relation is also corrective control.

The control question arises only when that dependency is embedded in a corrective process. Agreement-attraction errors show the perturbation clearly: speakers can be drawn toward a nearby mismatching noun and produce an agreement error (Bock & Miller, 1991). In online production monitoring, a speaker produces target forms, monitors the developing utterance, detects trouble, and repairs or recasts when the form fails to preserve the intended agreement relation (Levelt, 1983). On that loop, the tracked relation is agreement within the utterance, the lower-scale variables are particular target forms, and the response pathway is repair or recasting.

The agreement-specific character of the repair comes from the projected dependency itself. Once the controller's features and the domain are fixed, target forms are expected to express those values, so repair restores the higher-scale relation that made the profile projectible rather than only smoothing local fluency.

This keeps constraint and correction apart. A grammar may mark a mismatch as ill-formed without thereby supplying a corrective pathway. Corrective control requires the additional process by which a departure is detected and answered. The four marks may be met in the online repair loop: the mismatch threatens the projectible profile, agreement is the tracked relation, repair changes the lower-scale forms, and the higher-scale relation is preserved through that change.

Acquisition and normative correction are different candidate loops, not parts of a single system unless coordination is shown. In acquisition, learners converge on agreement patterns over developmental time; in normative contexts, teachers, editors, or interlocutors may reject or correct mismatches. These processes help stabilize agreement and may provide further control-like evidence, but they operate at different scales from online monitoring. The honest conclusion is conditional: agreement monitoring is not trivially homeostatic, but it supplies a specified linguistic loop rather than a loose analogy from grammatical dependency to feedback control.

Grammaticality at large is a harder and more contested candidate. If correction, repair, acquisition, transmission, and uptake together formed an architecture that returned usage to a community-sensitive profile, grammaticality could turn out to be control-secured, but whether grammaticality

is a single kind, and which loops would be doing the controlling, is just what is in dispute. The diagnostic leaves the possibility open without granting it, and agreement monitoring shows how a positive case would have to specify its loop.

13 WHAT THE DIAGNOSTIC CHANGES FOR HPC THEORY

The ladder matters for the wider HPC programme in linguistics and social ontology. Linguistic categories can have projection targets, stable profiles, network order, and maintainers without all of them being homeostatic in the strict control-theoretic sense, and social practices can be stabilized by records, offices, sanctions, training, and uptake without thereby constituting single homeostatic control systems. The diagnostic asks each case to display how far up the ladder it goes rather than inferring control from the mere fact of persistence.

A natural objection is that social correction and metalinguistic policing are themselves forms of feedback, and sometimes they are. The concession strengthens the diagnostic rather than weakening it, because it means that some linguistic and social categories may indeed be control-secured, but only where correction, repair, acquisition, transmission, and uptake combine into a coordinated architecture of the kind that agreement monitoring may exhibit. The point is that the evidence has to earn the upgrade case by case, rather than the label being granted on the strength of stability alone.

The distinction also bears on the more formal side of HPC work. It is tempting to operationalize homeostasis as a return-to-basin claim, asking whether a variable comes back to a set point after it has been perturbed. The governor and glycemic regulation both show why that test is too narrow, since what each system preserves is a relation, between workload and steam flow in the one case and between glucose availability and self-maintenance in the other, held in place through variation in the lower-scale variables.

A test that measures only the return of a variable to a fixed value can miss control that works precisely by allowing the variable to move. The basin is better defined over the relation the system preserves than over any set of fixed values.

For HPC work in linguistics this implies a division of labour. Many of the categories the framework would call homeostatic are, on inspection, stable projectible profiles, maintained profiles, or network-ordered kinds, which are real and worth a realist's defence without being control-secured. Reserving *HOMEOSTATIC* for the cases that display corrective control leaves the others untouched, since they remain real and projectible, while showing the analyst what to look for, and what would count as having found it, before the strongest description is reached for. What philosophers of science gain is a domain-neutral test that distinguishes projection target, stable profile, network order, maintenance, and corrective control, and that says what evidence would move a case from one rung to another.

14 CONCLUSION

Projectibility remains the point throughout. A stable cluster can underwrite projection without being homeostatic: what makes it projectible is the reliability of the profile for a specified inferential purpose, whereas what makes it homeostatic is a further corrective organization that preserves the relevant profile under perturbation. Drawn this way, the diagnostic protects Boyd's anti-essentialist insight while keeping *HOMEOSTATIC* from becoming a loose synonym for stable, useful, maintained, or causally organized.

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